Approximate minimax regret for conditionally similar tests

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Abstract

We construct approximately minimax regret optimal tests in a scalar Gaussian instrumental variables experiment. This example shows how conditionally similar tests allow a novel approximation approach, allowing us to optimize weights over alternatives to minimize worst-case power regret. Stochastic mirror descent and averaging yield finite-iteration guarantees with exact conditional similarity. We establish bounds for exact and simulated powers and illustrate the procedure numerically. This initial algorithm shows how existing nearly-optimal test computation methods can be extended to include composite alternative hypothesis sets without the computational cost associated with maintaining size control.

Keywords: Conditional similarity; minimax regret; weak instruments; stochastic mirror descent.
JEL classification: C12, C19.

1 Introduction

Conditional similarity controls rejection probabilities in the presence of nuisance parameters but generally leaves many tests to choose from (Moreira, 2003). We select among these tests by minimizing their largest power loss relative to the conditionally similar power envelope. This criterion is useful when instruments may be weak and no single test maximizes power at every alternative.

Our contribution is to use conditional size control to optimize power regret across alternatives. Elliott et al. (2015), henceforth EMW, fix a weighting distribution over alternatives; weighted average power then equals power against a single mixture distribution. Their numerical search concerns a least-favorable distribution under the null, and thus the algorithm seeks to find an optimal combination of tests that satisfy these size constraints simultaneously. While Section 4.3 of EMW does account for composite alternative hypothesis sets, our conditioning construction saves computing time, as we maintain size guarantees by construction. Aradillas Fernández et al. (2025) then develop stochastic mirror descent (SMD) for the corresponding finite-null, single-alternative problem, thus attaining approximation guarantees not present in the algorithm of EMW. That formulation also accommodates a fixed alternative mixture representing a weighted-average-power (WAP) criterion (for a discussion, see Section 3.8 of Lehmann and Romano (2005)), restricting the problem to that of a composite null and a singleton alternative.

We instead impose conditional similarity within every best response. This removes the need to optimize null weights to enforce size and allows the iterative search to focus on alternatives at which power falls furthest below its conditionally similar envelope. Aradillas Fernández et al. (2025) construct their algorithm

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using a subgradient of the Lagrangian, but by freeing up this size constraint, we are instead able to look at many combinations of alternative hypotheses. Weighted average power remains the best-response objective, but its weights are updated to solve a minimax regret problem rather than specified in advance as the final power criterion. Our result concerns the conditionally similar class in the Gaussian experiment, instead of the unrestricted class of level-α tests, but this flexible method may lend itself to tractable results beyond Gaussian settings, which is accomplished in Aradillas Fernández et al. (2025) for the single-alternative version of the problem.

For a finite alternative grid, we combine conditional Neyman–Pearson best responses with exponential weight updates, the latter of which help preserve the geometry of the probability simplex. Averaging preserves exact conditional similarity and yields explicit regret bounds for exact and simulated power calculations.

The contribution of this work lies in the conditionally similar testing literature. Andrews (2016) studies minimax regret within a class of conditional linear combination tests, while Chernozhukov et al. (2009) characterize admissible invariant similar tests and Montiel Olea (2020) characterizes admissible similar tests. Not yet established in the literature is a systematic way to compute minimax regret optimal tests, which is the main contribution of this work, as we optimize over the full conditionally similar class here. Furthermore, we connect to existing literature such as Andrews et al. (2006) that seeks to understand the power envelope for an invariant similar class exactly in the Gaussian example that we consider.

2 Setup and Algorithm

2.1 Testing problem

As in a weak instruments setting, we observe a pair of reduced-form estimates satisfying

$$\begin{pmatrix}\hat{\delta}\\ \hat{\pi}\end{pmatrix} \sim N\left(\begin{pmatrix}\pi\beta\\ \pi\end{pmatrix}, \Sigma\right), \qquad \Sigma \text{ positive definite,} \tag{1}$$

where Σ is known and does not depend on (β, π). We test $H_0 : \beta = \beta_0$ at level $\alpha \in (0, 1)$, with unrestricted nuisance parameter $\pi \in \mathbb{R}$. Define

$$S = \hat{\delta} - \beta_0\hat{\pi}, \qquad a = \mathrm{Cov}(\hat{\pi}, S)/\mathrm{Var}(S), \qquad U = \hat{\pi} - aS. \tag{2}$$

The transformation is invertible. Let $\sigma_S^2 = \mathrm{Var}(S)$ and $\sigma_U^2 = \mathrm{Var}(U)$. At $\theta = (\beta, \pi)$, the variables S and U are independent normals with means $m(\theta) = \pi(\beta - \beta_0)$ and $\tau(\theta) = \pi - am(\theta)$ and variances σ_S^2 and σ_U^2. Under the null, $S \mid U = u \sim N(0, \sigma_S^2)$, independently of π.

A randomized test is a measurable rejection probability $\varphi(s, u) \in [0, 1]$. Writing f_0 for the $N(0, \sigma_S^2)$ density, we consider

$$C_\alpha = \left\{\varphi : \int \varphi(s, u) f_0(s)\, ds = \alpha \text{ for Lebesgue-a.e. } u\right\}. \tag{3}$$

Because the null density of U is everywhere positive, this restriction is equivalent to conditional similarity for every π. It implies unconditional rejection probability α.

For alternatives $\theta \in \Theta \subseteq \{(\beta, \pi) : \beta \neq \beta_0\}$, define the power envelope, regret, and minimax value by

$$b(\theta) = \sup_{\varphi \in C_\alpha} \mathbb{E}_\theta[\varphi], \qquad \mathcal{R}(\varphi, \theta) = b(\theta) - \mathbb{E}_\theta[\varphi], \qquad V_\Theta = \inf_{\varphi \in C_\alpha} \sup_{\theta \in \Theta} \mathcal{R}(\varphi, \theta). \tag{4}$$

We first take $\Theta = \Theta_I = \{\theta_1, \ldots, \theta_I\}$, with $I \geq 2$, and write $V_I = V_{\Theta_I}$ and $b_i = b(\theta_i)$.

2.2 Conditional best responses and weight updates

For weights w in the probability simplex Δ^{I-1}, let $g_w = \sum_i w_i g_i$, where g_i is the joint density of (S, U) at θ_i. The best response maximizes weighted average power:

$$\varphi_w \in \arg\max_{\varphi \in C_\alpha} \sum_{i=1}^{I} w_i \mathbb{E}_{\theta_i}[\varphi]. \tag{5}$$

The weights w are optimization variables for worst-case regret. Thus each best response uses a single alternative mixture, but the mixture changes across iterations as the algorithm reallocates weight toward high-regret alternatives. The constraint in (3) separates across u. Conditional Neyman–Pearson optimization therefore rejects for large values of

$$L_w(s,u) = \frac{g_w(s,u)}{f_0(s)} = \sum_{i=1}^{I} w_i f_U(u;\tau_i) \exp\left(\frac{m_i s}{\sigma_S^2} - \frac{m_i^2}{2\sigma_S^2}\right), \tag{6}$$

where $m_i = m(\theta_i)$, $\tau_i = \tau(\theta_i)$, and $f_U(\cdot;\tau_i)$ is the $N(\tau_i, \sigma_U^2)$ density. Specifically, $\varphi_w = \mathbf{1}\{L_w > c_w(u)\} + \gamma_w(u)\mathbf{1}\{L_w = c_w(u)\}$, with threshold and boundary randomization chosen to satisfy (3). For a single alternative, the optimal rule is one-sided in S, giving

$$b_i = \Phi\Big(\Phi^{-1}(\alpha) + |m_i|/\sigma_S\Big), \tag{7}$$

where Φ is the standard normal CDF. When $m_i = 0$, the envelope equals α.

The function $F(w) = \sup_{\varphi \in C_\alpha} \sum_i w_i(\mathbb{E}_{\theta_i}[\varphi] - b_i)$ is convex, with subgradient $G_i(w) = \mathbb{E}_{\theta_i}[\varphi_w] - b_i$. Negative-entropy mirror descent thus increases weights in proportion to estimated regret (see Section 4.2 of Bubeck (2015), as well as Freund and Schapire (1999), after which this algorithm is modeled). Starting from $w_{1,i} = 1/I$, at iteration k we compute $\varphi_k = \varphi_{w_k}$ and set

$$\widehat{p}_{k,i} = \frac{1}{N}\sum_{n=1}^{N} \varphi_k(Y_{k,i,n}), \qquad \widehat{G}_{k,i} = \widehat{p}_{k,i} - b_i, \qquad w_{k+1,i} = \frac{w_{k,i}\exp(-\eta \widehat{G}_{k,i})}{\sum_j w_{k,j}\exp(-\eta \widehat{G}_{k,j})}. \tag{8}$$

Conditional on the preceding iterations, the draws $Y_{k,i,n} \sim g_i$ are independent across i and n and fresh at each iteration. Hence $\widehat{G}_k$ is conditionally unbiased. Exact-power updates replace $\widehat{p}_{k,i}$ by $\mathbb{E}_{\theta_i}\varphi_k$.

We return $\bar{\varphi}_K = K^{-1}\sum_{k=1}^{K}\varphi_k$, which could also be implemented by selecting an iteration uniformly and applying its test. Note that, conditional on simulation history, $\bar{\varphi}_K \in C_\alpha$ by construction. Thus simulation affects regret but does not affect conditional size when best responses are computed exactly. Algorithm 1 specifies the full implementation.

2.3 Algorithm guarantees

Proposition 1. *For the exact conditional best responses in* (5)*, the following statements hold for every $K \geq 1$ and $\eta > 0$.*

1. *With exact-power updates,*

$$\max_i \mathcal{R}(\bar{\varphi}_K, \theta_i) \leq V_I + \frac{\log I}{\eta K} + \frac{\eta}{8}. \tag{9}$$

2. *With the simulated updates in* (8)*, for any $\delta \in (0,1)$, with probability at least $1-\delta$ over the simulation draws,*

$$\max_i \mathcal{R}(\bar{\varphi}_K, \theta_i) \leq V_I + \frac{\log I}{\eta K} + \frac{\eta}{2} + \sqrt{\frac{2\log((I+1)/\delta)}{NK}}. \tag{10}$$

Algorithm 1 Conditionally similar testing by SMD

Require: Grid Θ_I, size α, tolerance ϵ, failure probability δ, draws N.

Choose K and η according to (11).
Compute envelopes b_i using (7); set $w_{1,i} = 1/I$.
for $k = 1, \ldots, K$ **do**
    Construct φ_k from (6), imposing conditional size α.
    Draw fresh $Y_{k,i,n} \sim g_i$ independently for all i, n.
    Set $\widehat{p}_{k,i} = N^{-1} \sum_n \varphi_k(Y_{k,i,n})$ for every i.
    Set $w_{k+1,i} \propto w_{k,i} \exp\{\eta(b_i - \widehat{p}_{k,i})\}$; normalize.
end for
return $\bar{\varphi}_K = K^{-1} \sum_k \varphi_k$.

With exact powers, $\eta = \sqrt{8 \log I/K}$ gives error $\sqrt{\log I/(2K)}$. With simulated powers, an error tolerance $\epsilon > 0$ and failure probability δ are attained by

$$K = \left\lceil \epsilon^{-2} \left(\sqrt{2 \log I} + \sqrt{\frac{2 \log((I+1)/\delta)}{N}} \right)^2 \right\rceil, \qquad \eta = \sqrt{\frac{2 \log I}{K}}. \tag{11}$$

The guarantees concern the average test, not the best response to average weights, which may underperform (e.g., see the discussion surrounding Algorithm 1 of Aradillas Fernández et al. (2025)). Note that these results do not require a minimax interchange; weighted regret is bounded by V_I directly. Indeed, (A.1) also establishes $V_I = -\min_{w \in \Delta^{I-1}} F(w)$ by letting K increase, combining weak duality with the bound on average weighted regret. The continuity of F ensures attainment.

3 A numerical illustration

We set $\beta_0 = 0$, $\Sigma = I_2$, and $\alpha = 0.05$, so that $S = \hat{\delta}$ and $U = \hat{\pi}$. Our grid contains $\theta_1 = (-3, 0.5)$ and $\theta_2 = (1, 2)$, giving $(m_1, m_2) = (-1.5, 2)$ and $(\tau_1, \tau_2) = (0.5, 2)$. The corresponding power envelopes are 0.4424 and 0.6388. This deliberately small grid isolates the trade-off between power against negative and positive effects.

For these two alternatives, the likelihood ratio in (6) is strictly convex in s. The best response rejects in two tails. If $q_w(u)$ is its left-tail null probability, the critical values satisfy

$$\ell_w(u) = \Phi^{-1}(q_w(u)), \qquad h_w(u) = \Phi^{-1}(1 - \alpha + q_w(u)), \qquad L_w(\ell_w(u), u) = L_w(h_w(u), u). \tag{12}$$

We solve the last equality by bisection over $q \in [10^{-14}, \alpha - 10^{-14}]$. The resulting rule has conditional size $q + (\alpha - q) = \alpha$ by construction. Larger values of u favor the second alternative and shift rejection probability toward the right tail.

We set the target excess regret to $\epsilon = 0.02$, the failure probability to $\delta = 0.05$, and the number of draws per alternative per iteration to $N = 512$. Equation (11), with $I = 2$, then selects $K = 4{,}251$ and $\eta = 0.01806$. Under exact conditional best responses, Proposition 1 gives probability at least 0.95 that the average test's maximum grid regret is at most $V_I + 0.02$. We evaluate power by integrating the conditional rejection probability with 320-node Gauss–Hermite quadrature. Doubling the quadrature order from 160 to 320 changes computed iteration-specific powers by less than 10^{-12}, suggesting numerical error is very low here. The reported values then come from one simulation run.

Table 1 shows that conditioning improves power at both alternatives relative to the equal-tail test. Figure 2 shows how the conditioning statistic shifts rejection probability between tails. Subsequent weight updates

Table 1: Power and regret on the two-alternative grid

Test	Power at θ_1	Power at θ_2	Max. regret
Equal-tail test	0.323	0.516	0.123
Initial best response	0.361	0.567	0.082
SMD average, 100 iterations	0.361	0.567	0.081
SMD average, 4251 iterations	0.365	0.563	0.078

Notes: Size for all cases is $\alpha = .05$. The equal-tail test rejects when $|S| > \Phi^{-1}(0.975)$. The initial best response uses $w = (1/2, 1/2)$. SMD rows average all tests through the indicated iteration of the same run. Powers are evaluated by numerical integration; size follows from the tail allocation. Maximum regret is relative to the pointwise envelopes in (7).

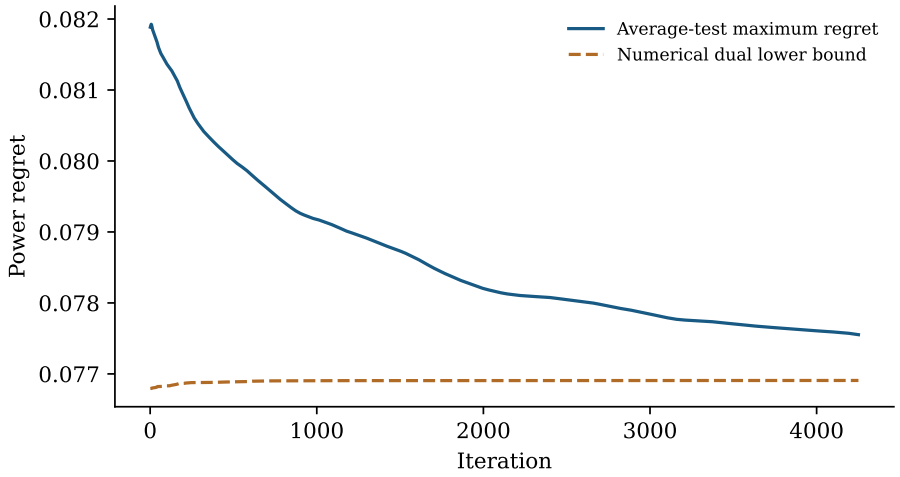


Figure 1: Regret along the SMD iterations. We report the maximum regret of the average test and the running maximum of $w_k^\top r_k$, evaluated by numerical integration. With exact best responses and integration, the latter bounds V_I from below. OpenAI Codex assisted in preparing the Python code used to generate this figure.

increase power at θ_1 while sacrificing some power at θ_2, reducing the largest regret. After 4,251 iterations, maximum regret is 0.0776, compared with 0.0819 initially and 0.1228 for the equal-tail test. Figure 1 reports a numerical dual lower bound of 0.0769, leaving a numerical gap of 0.00065. The small distance between the initial best response and the output of Algorithm 1 suggests the example here could benefit from early stopping, but recall that Proposition 1 is only a worst-case guarantee. Furthermore, monotonicity in average-test maximum regret is not always guaranteed, and in fact between the first and second iterations, regret goes up in our example. The dual lower bound therefore only acts as a guide on the maximum amount by which minimax regret can be reduced.

4 Conclusion

Conditional similarity allows us to shift the numerical search from null distributions enforcing size to alternative distributions determining worst-case power regret. Relative to EMW's fixed WAP criterion, the alternative weights are chosen endogenously to balance regret within the conditionally similar class, allowing robustness to WAP misspecification. Conditional Neyman–Pearson best responses and SMD deliver approximation guarantees while preserving exact conditional size. This work can serve as the basis for future work moving beyond the Gaussian setup of the problem, with the built-in size control allowing for optimization over many alternative hypotheses.

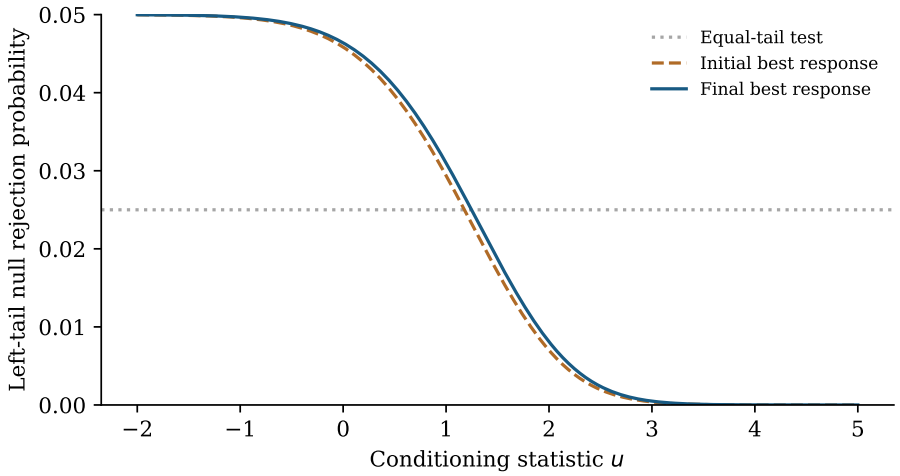


Figure 2: Conditional tail allocation. Left-tail null rejection probabilities for the initial and final individual best responses are compared with the equal-tail test. OpenAI Codex assisted in preparing the Python code used to generate this figure.

Data and code availability

We use simulated data. The reproducible Python package, as well as instructions, are available at `https://github.com/aaf101/SMD-for-Conditional-Similarity`.

A Proofs of Main Results

A.1 Proof of Proposition 1

This proof follows a standard Hedge derivation. Write $r_{k,i} = \mathcal{R}(\varphi_k, \theta_i) \in [0,1]$. Best-response optimality gives $w_k^\top r_k = \inf_{\varphi \in C_\alpha} \sum_i w_{k,i} \mathcal{R}(\varphi, \theta_i) \le V_I$. The exponential-weights inequality for rewards in $[0,1]$ (Theorem 2.2 of Cesa-Bianchi and Lugosi (2006)) yields

$$\max_i \frac{1}{K} \sum_k r_{k,i} \le \frac{1}{K} \sum_k w_k^\top r_k + \frac{\log I}{\eta K} + \frac{\eta}{8}. \tag{A.1}$$

Linearity of power proves (9).

For simulated updates, $\widehat{G}_{k,i} \in [-1,1]$. The same inequality, now with range length two, gives, simultaneously for all i,

$$\frac{1}{K} \sum_k (r_{k,i} - w_k^\top r_k) \le \frac{\log I}{\eta K} + \frac{\eta}{2} + \frac{1}{K} \sum_k \Delta_{k,i} - \frac{1}{K} \sum_k w_k^\top \Delta_k, \tag{A.2}$$

where $\Delta_{k,i} = \widehat{G}_{k,i} - G_i(w_k)$. Conditional Hoeffding bounds and iteration over the simulation history imply $\Pr(K^{-1} \sum_k \Delta_{k,i} > x) \le \exp(-2NKx^2)$. The same bound holds for $-K^{-1} \sum_k w_k^\top \Delta_k$: conditional independence gives squared summand ranges totaling $\sum_i w_{k,i}^2 / N \le 1/N$ at each iteration. A union bound over these $I+1$ deviations, with $x = \sqrt{\log((I+1)/\delta)/(2NK)}$, bounds the two noise terms in (A.2) by $2x$. Using $w_k^\top r_k \le V_I$ and linearity proves (10).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used GPT-6 through OpenAI Codex to help with the writing, proof-checking, code implementation, and preparation of the numerical illustration and figures in this paper. The author directed this work and subsequently reviewed and verified all output. The author takes full responsibility for the content of the article.

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